

Windows Subsystem for Linux (WSL)

Intro. to Computing

<https://mandarm.github.io/computing.html>

Installation

- Open Windows PowerShell (WPS).
- Run `wsl --install`.
 - If/when it says Downloading: Ubuntu, type Control-C to abort.
 - Otherwise, let it install the Virtual Machine Platform.
- Run `wsl --update`.
(should say most recent version is already installed)
- If you do not already have a copy of
<https://releases.ubuntu.com/26.04/ubuntu-26.04-wsl-amd64.wsl>,
download it.
- Double-click the file to launch the installer, OR run
`wsl --install --name ubuntu26 --from-file .\ubuntu-26.04-wsl-amd64.wsl`
- Close the window or type Control-D to exit.
- Run `wsl` in WPS to start WSL again.
- Run `sudo apt update && sudo apt upgrade`.

■ How do Windows locations appear in WSL and vice versa?

- `C:\` $\equiv$ `/mnt/c`
- `/` $\equiv$ `\\wsl.localhost\Ubuntu-26.04` or `\\wsl$` (in Windows File explorer address bar)

■ **General guideline:** for best performance, avoid working across environments (Windows and WSL)

- When programming in WSL, store files under `/home/student` (or `/home/cssc ...`)
not `/mnt/c/Users/CSSC/Project` or `C:\Users\CSSC\Project`
- Exception (for now): edit your Python programs using Windows; run them using WSL.

■ **Warning re: file/directory names:**

Windows $\rightarrow$ case-insensitive, Linux $\rightarrow$ case-sensitive

Mixing Windows and WSL commands

- Using Windows executables from WSL: include .exe extension

Examples:

```
explorer.exe .
```

```
notepad.exe .bashrc
```

```
notepad.exe .bash_aliases
```

```
notepad.exe "C:\temp\foo.txt"
```

```
notepad.exe C:\\temp\\foo.txt
```

BUT `notepad.exe ~/.bash_aliases` **did not work for me!**

- Using Linux programs from Windows: add `wsl` before command name

Examples:

```
wsl ls -l
```

My personal experience: does not always work as expected.

Searching / installing packages

1. Update list of available packages: `sudo apt update`
(you only need to do this occasionally, or if things don't work as expected)
2. Search for the package name if necessary. Examples:
`apt-cache search less` `apt-cache search gnome-terminal`
3. Install the package: `sudo apt install <package name>`

Installing useful applications

- **Terminal:** `sudo apt install gnome-terminal -y`
- **Simple text editor:** (for quick edits) `sudo apt install gedit -y`
OR `sudo apt install gnome-text-editor -y`
- **Python**
 - check python version by running `python3 --version`
 - if you are not running Python 3.14
 - `sudo apt install software-properties-common`
 - `sudo add-apt-repository ppa:deadsnakes/ppa`
 - `sudo apt update`
 - `sudo apt install python3.14 python3.14-venv`

Integrating with VS Code

- **Overview:** learn.microsoft.com/en-us/windows/wsl/about
- **All docs:** learn.microsoft.com/en-us/windows/wsl
- **Running GUI apps** (e.g., a simple editor, *meld* — program for viewing differences between two similar files)
learn.microsoft.com/en-us/windows/wsl/tutorials/gui-apps
- **Visual Studio Code**
learn.microsoft.com/en-us/training/modules/developing-in-wsl